

# Getting into genes:

## Summary

### Answers

Use the listed words to complete the sentences below.

|           |             |          |           |          |           |
|-----------|-------------|----------|-----------|----------|-----------|
| amino     | chromosomes | diploid  | dominant  | errors   | forensic  |
| gametes   | generation  | genetics | genotypes | guanine  | helix     |
| identical | meiosis     | nitrogen | opposite  | physical | replicate |
| two       |             |          |           |          |           |

- The chromosomes of cells contain genes made of DNA.
- Genes determine many physical characteristics (e.g. eye colour).
- Diploid human cells contain 23 chromosomal pairs. Gametes cells, however, contain only 23 chromosomes.
- In a DNA molecule the amounts of thymine and adenine bases are always the same, and the amount of cytosine and guanine bases are always the same.
- The Watson–Crick DNA model is a double helix like a twisted ladder.
- Each step of the DNA helix is made up of two nitrogen bases and the bases are attached to the sugar units that are part of the sides of the ladder.
- The steps of the DNA helix are composed of paired nitrogen bases. There is always an adenine base opposite a thymine base and a cytosine base opposite a guanine base.
- Polypeptides are made up of units called amino acids. A set of three DNA bases codes for a particular amino acid.
- During cell division (mitosis) the DNA needs to replicate to ensure the new cells have identical copies of the DNA.
- In most instances the DNA is copied exactly during mitosis. In some cases errors occur and mutations result.
- Gametes are formed by a type of cell division called meiosis.
- The fertilised egg has a diploid number of chromosomes.
- X and Y are the sex chromosomes. Females have two X chromosomes and males have an X and a Y chromosome.
- Different forms of a gene are called alleles. Alleles can be dominant or recessive. Two copies of a recessive gene are required for the characteristic to be expressed.
- Gregor Mendel was an Austrian monk who conducted the first quantitative genetics experiments using garden peas.
- Punnett squares are a useful tool to predict the proportion of genotypes and phenotypes in the offspring.
- Pedigree diagrams are useful in showing how various genes are transferred from one generation to another.
- DNA fingerprinting is used in forensic investigations and paternity tests.
- Clones are organisms that are genetically identical.